

Determinant Divisibility of Centered Latin Squares: A Unified Theorem and the $\mathbb{F}_2$ -Rank Obstruction

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Abstract

Let L be an $n \times n$ Latin square with entries in $\{1, \dots, n\}$ and let $E_{\text{std}} = P^T E P$ denote the Gram-projected $(n-1) \times (n-1)$ matrix of the centered operator $E = L - \frac{n+1}{2} J$ on the standard subspace V_{std} , where P is the basis matrix of V_{std} .

We prove three main results. First, $\frac{n^2}{\gcd(n, 2)}$ divides $\det(E_{\text{std}})$ for every Latin square of every order $n \geq 2$; for odd n the stronger bound $n^2 \mid \det(E_{\text{std}})$ holds universally (*unified divisibility*). Second, for even n we give a complete $\mathbb{F}_2$ -rank characterization of the Latin squares that achieve the improved divisibility $n^2 \mid \det(E_{\text{std}})$: when $n \equiv 2 \pmod{4}$ the criterion is $\text{rank}_{\mathbb{F}_2}(A \bmod 2) < n-1$; when $n \equiv 0 \pmod{4}$ the criterion is $\text{adj}(A \bmod 2) \cdot \mathbf{1} = \mathbf{0}$ over $\mathbb{F}_2$. In both cases the parity pattern $L \bmod 2$ is a complete invariant (*complete binary criterion*). Equivalently, the obstruction to n^2 -divisibility is entirely 2-adic: every odd prime factor of n contributes its share unconditionally. Third, we prove that counterexamples to n^2 -divisibility exist for every $n \equiv 2 \pmod{4}$, $n \geq 6$, via an explicit infinite family of skip-one circulant parity patterns (*universal counterexamples*).

These results are complemented by p -adic lower bounds: $v_p(\det A) \geq v_p(n) + \max(0, k_p - 1)$ for every odd prime $p \mid n$, and $v_2(\det A) \geq v_2(n/2) + \max(0, k_2 - 1)$ when n is even (empirically tight across sampled instances up to $n \leq 20$), a cokernel computation for cyclic Latin squares, and exhaustive/constructive counterexample analyses at orders 6 and 10. The determinant side of the story may be viewed as complementary to earlier work on determinants and permanents of Latin squares [7, 3].

Keywords: Latin squares, determinant divisibility, $\mathbb{F}_2$ -rank, centered matrix, constant-weight code, parity pattern.

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1 Introduction

A *Latin square* of order n is an $n \times n$ array $L = (L_{ij})$ whose entries belong to $\{1, \dots, n\}$ and in which every row and every column is a permutation of $\{1, \dots, n\}$. Latin squares are fundamental objects in combinatorics with deep connections to algebra, coding theory, and experimental design [5, 14]. Determinant questions for Latin squares have also been studied from other viewpoints, including order-specific determinant calculations and determinant / permanent polynomials [7, 3].

Given a Latin square L , the *centered matrix* $E = L - \frac{n+1}{2} J$ (where J is the all-ones matrix) satisfies $E\mathbf{1} = \mathbf{0}$ and $\mathbf{1}^T E = \mathbf{0}$, so $\text{rank}(E) \leq n-1$ and E acts naturally on the hyperplane $V_{\text{std}} = \{x \in \mathbb{R}^n : \sum x_i = 0\}$. The Gram-projected matrix $E_{\text{std}} = P^T E P$ (where P is the basis matrix of V_{std} ; see Section 2) carries algebraic information about the Latin square that is invisible to purely combinatorial invariants.

In this paper we study the arithmetic of $\det(E_{\text{std}})$. The main contributions are threefold.

- (a) *Unified divisibility.* For every Latin square of order $n \geq 2$,

$$\frac{n^2}{\gcd(n, 2)} \mid \det(E_{\text{std}}).$$

In particular, if n is odd then the stronger divisibility $n^2 \mid \det(E_{\text{std}})$ holds universally (Corollary 3.3).

- (b) *Complete binary criterion.* The obstruction to n^2 -divisibility is entirely 2-adic: every odd prime factor of n contributes its share unconditionally. When $n \equiv 2 \pmod{4}$, the criterion is rank-theoretic:

$$n^2 \mid \det(E_{\text{std}}) \iff \text{rank}_{\mathbb{F}_2}(A \bmod 2) < n - 1.$$

When $n \equiv 0 \pmod{4}$, rank alone is not sufficient; the complete criterion is the adjugate condition

$$\text{adj}(A \bmod 2) \cdot \mathbf{1} = \mathbf{0} \quad \text{in } \mathbb{F}_2^{n-1}.$$

In both congruence classes, the parity pattern $L \bmod 2$ is a complete invariant for the n^2 -divisibility question.

- (c) *Universal counterexamples.* For every $n \equiv 2 \pmod{4}$, $n \geq 6$, there exist Latin squares violating n^2 -divisibility (Theorem 4.14). Concrete examples at $n = 6$ (exhaustive) and $n = 10$ (constructed) are given in Section 7.

Secondary results include the constant row-weight theorem for $A \bmod 2$ (Theorem 5.1), the balanced realizability proposition (Proposition 4.12), the sharp p -adic bounds via Smith normal form (Theorem 6.1 and corollary 6.3), and the cokernel computation for cyclic Latin squares (Corollary 3.9).

The proof rests on two independent mechanisms. The first is a geometric factor of n arising from the Gram matrix of the standard basis for V_{std} . The second is an arithmetic factor: the row sums of A are always divisible by n (when n is odd) or by $n/2$ (when n is even), which forces $\det(A)$ to acquire a corresponding factor via the adjugate identity. These two factors combine to yield the quadratic bound.

The paper is organized as follows. Section 2 fixes notation and establishes the Gram factorization. Section 3 proves the unified divisibility theorem. Section 4 establishes the binary criterion: the rank characterization for $n \equiv 2 \pmod{4}$, the adjugate criterion for $n \equiv 0 \pmod{4}$, and the universality of counterexamples for all $n \equiv 2 \pmod{4}$. Section 5 derives structural properties of $A \bmod 2$. Section 6 develops the Smith-normal-form and p -adic bounds. Section 7 presents the counterexamples. Section 8 summarizes the computational methods. Section 9 concludes with a summary, related directions, and open problems.

2 Preliminaries and notation

Throughout, L is an $n \times n$ Latin square with entries in $\{1, \dots, n\}$, indexed by $0 \leq i, j \leq n - 1$. For a prime p and a nonzero integer m , we write $v_p(m)$ for the p -adic valuation; by convention, $v_p(0) = +\infty$. All p -adic lower bounds in this paper are therefore vacuously satisfied when $\det(A) = 0$ (which can occur, e.g. at $n = 4$).

Definition 2.1 (Centered matrix). Set $m = \frac{n+1}{2}$ and $E = L - mJ$. The entries of E are half-integers when n is even and integers when n is odd.

Since every row and column of L sums to $\frac{n(n+1)}{2} = nm$, we have $E\mathbf{1} = \mathbf{0}$ and $\mathbf{1}^T E = \mathbf{0}$.

Definition 2.2 (Standard representation). Let $V_{\text{std}} = \ker(\mathbf{1}^T) \subset \mathbb{R}^n$ and define the basis $b_i = e_i - e_{n-1}$ for $0 \leq i \leq n - 2$, where e_i is the i -th standard unit vector. The *Gram matrix* is

$$G = P^T P, \quad P = [b_0 \mid \dots \mid b_{n-2}] \in \mathbb{R}^{n \times (n-1)},$$

so $G = I_{n-1} + J_{n-1}$ and $\det(G) = n$.

Definition 2.3 (Difference matrix). Define the $(n-1) \times (n-1)$ integer matrix

$$A_{ij} = L_{ij} - L_{i,n-1}, \quad 0 \leq i, j \leq n-2.$$

Lemma 2.4 (Gram factorization). $E_{\text{std}} = G \cdot A$, so $\det(E_{\text{std}}) = n \cdot \det(A)$.

Proof. Since $E\mathbf{1} = \mathbf{0}$, the operator E maps V_{std} into itself. Writing $EP = PA$, the first $n-1$ rows are verified entry-wise by

$$(EP)_{k,j} = E_{k,j} - E_{k,n-1} = L_{kj} - L_{k,n-1} = A_{kj}$$

for $0 \leq k, j \leq n-2$, since the m -terms cancel. For the last row, use $\mathbf{1}^T E = \mathbf{0}^T$. For each $0 \leq j \leq n-2$,

$$(EP)_{n-1,j} = E_{n-1,j} - E_{n-1,n-1} = -\sum_{r=0}^{n-2} (E_{rj} - E_{r,n-1}) = -\sum_{r=0}^{n-2} A_{rj}.$$

This is exactly the last row of PA , since the last row of P is $(-1, \dots, -1)$. Thus $EP = PA$, and so we obtain $E_{\text{std}} = P^T EP = P^T PA = GA$, whence $\det(E_{\text{std}}) = \det(G) \det(A) = n \det(A)$. $\square$

Remark 2.5. The factorization $E_{\text{std}} = GA$ distinguishes two objects:

- A is the coordinate matrix of $E|_{V_{\text{std}}}$ in the basis $\{b_i\}$: it records the action $Eb_j = \sum_i A_{ij} b_i$.
- $E_{\text{std}} = GA$ is the Gram-projected matrix, with entries $(E_{\text{std}})_{ij} = \langle b_i, Eb_j \rangle = (P^T EP)_{ij}$.

In particular, $\det(E_{\text{std}}) = n \cdot \det(A)$, where the factor $n = \det(G)$ reflects the non-orthonormality of $\{b_i\}$. The invariant $\det(A) \in \mathbb{Z}$ is the determinant of the operator $E|_{V_{\text{std}}}$; $\det(E_{\text{std}})$ absorbs an additional geometric factor. Both are studied in this paper; most divisibility statements are formulated for $\det(E_{\text{std}})$ and then equivalently expressed in terms of $\det(A)$.

3 The unified divisibility theorem

Lemma 3.1 (Row-sum identity). For every $0 \leq i \leq n-2$,

$$\sum_{j=0}^{n-2} A_{ij} = \frac{n}{2}(n+1 - 2L_{i,n-1}).$$

Write $c_i = n+1 - 2L_{i,n-1}$.

- If n is even, then c_i is odd and $A\mathbf{1} = \frac{n}{2} \mathbf{c}$.
- If n is odd, then c_i is even; writing $u_i = c_i/2 \in \mathbb{Z}$, one has $A\mathbf{1} = n \mathbf{u}$.

Proof. We compute

$$\begin{aligned} \sum_{j=0}^{n-2} A_{ij} &= \sum_{j=0}^{n-2} (L_{ij} - L_{i,n-1}) = \left(\sum_{j=0}^{n-1} L_{ij} \right) - L_{i,n-1} - (n-1)L_{i,n-1} \\ &= \frac{n(n+1)}{2} - nL_{i,n-1} = \frac{n}{2}(n+1 - 2L_{i,n-1}). \end{aligned}$$

Since $2L_{i,n-1}$ is always even: if n is even then $n+1$ is odd, so c_i is odd; if n is odd then $n+1$ is even, so c_i is even and $u_i = c_i/2 = (n+1 - 2L_{i,n-1})/2$ is an integer. $\square$

Theorem 3.2 (Unified divisibility). For every Latin square L of order $n \geq 2$,

$$\frac{n^2}{\gcd(n, 2)} \mid \det(E_{\text{std}}).$$

Proof. By Lemma 2.4, $\det(E_{\text{std}}) = n \det(A)$. It suffices to show that $\frac{n}{\gcd(n,2)} \mid \det(A)$. We treat odd and even n separately.

Case 1: n odd. By Lemma 3.1, every row sum of A is $\frac{n}{2}(n+1-2L_{i,n-1})$. Since $n+1$ is even when n is odd, $(n+1-2L_{i,n-1})/2$ is an integer, and so each row sum equals $n \cdot (\frac{n+1-2L_{i,n-1}}{2})$. Hence $A\mathbf{1} \equiv \mathbf{0} \pmod{n}$. By the adjugate identity $\text{adj}(A) \cdot A\mathbf{1} = \det(A) \cdot \mathbf{1}$, it follows that $n \mid \det(A)$. Combined with $\det(E_{\text{std}}) = n \det(A)$, we obtain $n^2 \mid \det(E_{\text{std}})$.

Case 2: n even. Set $h = n/2$. By Lemma 3.1, every row sum equals $h \cdot (\text{odd})$, so $A\mathbf{1} \equiv \mathbf{0} \pmod{h}$. By the adjugate identity, $h \mid \det(A)$. Therefore $\det(E_{\text{std}}) = n \det(A)$ is divisible by $n \cdot h = n^2/2 = n^2/\gcd(n,2)$. $\square$

Corollary 3.3. If n is odd, then $n^2 \mid \det(E_{\text{std}})$.

Proof. This is Case 1 of Theorem 3.2: each row sum of A is divisible by n , so $n \mid \det(A)$ and $\det(E_{\text{std}}) = n \det(A)$ is divisible by n^2 . $\square$

Remark 3.4 (Prime-by-prime decomposition). Let p be an odd prime dividing n . By Lemma 3.1, each row sum of A is divisible by n when n is odd (since $A\mathbf{1} = n\mathbf{u}$) and by $n/2$ when n is even (since $A\mathbf{1} = \frac{n}{2}\mathbf{c}$ with c_i odd). In either case $v_p(n) \leq v_p(\text{row sum})$, so $A\mathbf{1} \equiv \mathbf{0} \pmod{p^{v_p(n)}}$ and therefore $p^{v_p(n)} \mid \det(A)$.

Proposition 3.5 (Sharpness). The bound in Theorem 3.2 is sharp for $n \in \{2, 6, 8, 10\}$. Specifically, there exist Latin squares satisfying $\frac{n^2}{2} \mid \det(E_{\text{std}})$ but $n^2 \nmid \det(E_{\text{std}})$ at each of these orders.

- (i) For $n = 2$, $\det(E_{\text{std}}) = -2$ and $4 \nmid -2$.
- (ii) For $n = 6$, there are exactly 576 reduced Latin squares with $18 \mid \det(E_{\text{std}})$ but $36 \nmid \det(E_{\text{std}})$ (Proposition 7.1).
- (iii) For $n = 8$, explicit Jacobson–Matthews samples satisfy $32 \mid \det(E_{\text{std}})$ but $64 \nmid \det(E_{\text{std}})$; all such squares have $\dim \ker_{\mathbb{F}_2}(A \bmod 2) = 1$, explained by the adjugate criterion (Theorem 4.7).
- (iv) For $n = 10$, explicit Latin squares satisfy $50 \mid \det(E_{\text{std}})$ but $100 \nmid \det(E_{\text{std}})$ (Proposition 7.5).

Remark 3.6. At $n = 4$, exhaustive computation over all 576 Latin squares shows that $n^2 = 16$ divides $\det(E_{\text{std}})$ universally; the weaker bound $n^2/2 = 8$ is therefore *not* sharp at this order. Whether $n = 4$ is the only even order where the bound can be strengthened remains open.

Proof. Items (i), (ii), and (iv) are exhibited in Section 7. Item (iii): a sharpness witness at $n = 8$ is the Latin square obtained from the first Jacobson–Matthews sample (with $5n^2 = 320$ mixing steps, seed = 0) satisfying $\dim \ker_{\mathbb{F}_2} = 1$; its determinant is verified by Bareiss elimination to satisfy $32 \mid \det(E_{\text{std}})$ but $64 \nmid \det(E_{\text{std}})$. $\square$

Remark 3.7. In a corpus of 10 000 Jacobson–Matthews samples at $n = 8$ ($5n^2$ mixing steps each), approximately 11% satisfy $32 \mid \det(E_{\text{std}})$ but $64 \nmid \det(E_{\text{std}})$. This sampling frequency is not claimed as a rigorous proportion of all Latin squares of order 8.

Proposition 3.8 (Cyclic formula). For the cyclic Latin square $L_{ij} = (i + j) \bmod n + 1$ of any order $n \geq 2$:

$$\det(E_{\text{std}})_{\text{cyc}} = (-1)^{\lfloor n/2 \rfloor} \cdot n^{n-1}.$$

In particular, $|\det(E_{\text{std}})| = n^{n-1}$, i.e. the exact exponent of n dividing $\det(E_{\text{std}})$ is $n-1$. (Singular Latin squares with $\det(A) = 0$ exist already at $n = 4$, so no absolute optimality claim is intended here.)

Proof. The centered matrix E has entries $E_{ij} = (i+j) \bmod n - (n-1)/2 =: h((i+j) \bmod n)$ and is real symmetric since $E_{ij} = E_{ji}$. Because $E\mathbf{1} = \mathbf{0}$, the eigenvalues of A are all real (since $A = G^{-1}E_{\text{std}}$ is similar to the real symmetric matrix $G^{-1/2}E_{\text{std}}G^{-1/2}$), and $\det(A) = \prod \mu_k$.

Since E_{ij} depends only on $(i+j) \bmod n$ (a reverse-circulant or Hankel-circulant structure), the DFT vectors satisfy $Ev^{(k)} = \hat{h}(k)v^{(n-k)}$ for each $k = 1, \dots, n-1$, where $v_j^{(k)} = \omega^{jk}$, $\omega = e^{2\pi i/n}$, and $\hat{h}(k) = \sum_{j=0}^{n-1} h(j)\omega^{jk}$. (The reversal $v^{(k)} \mapsto v^{(n-k)}$ arises because $(Mv^{(k)})_i = \sum_j h(i+j)\omega^{jk} = \omega^{-ik}\hat{h}(k) = \hat{h}(k)v_i^{(n-k)}$.) Evaluating explicitly: $h(j) = j - (n-1)/2$ for $j = 0, \dots, n-1$ gives $\hat{h}(k) = n/(\omega^k - 1)$ for $k \neq 0$, since $\sum_{j=0}^{n-1} j\omega^{jk} = -n/(\omega^k - 1)$ and $\sum_j \omega^{jk} = 0$. Writing $\hat{h}(k) = a_k + ib_k$ and passing to the real basis $u_k = \Re v^{(k)}$, $w_k = \Im v^{(k)}$, one finds (see Davis [4], Chap. 3 for the general spectral theory of circulant and Hankel-circulant matrices) that E acts on $\langle u_k, w_k \rangle$ by the matrix $\begin{bmatrix} a_k & b_k \\ b_k & -a_k \end{bmatrix}$, whose eigenvalues are $\pm|\hat{h}(k)| = \pm n/|\omega^k - 1|$ —one positive and one negative. Since $v^{(n-k)} = \overline{v^{(k)}}$, the pair $\{k, n-k\}$ shares the same real subspace $W_k = \langle u_k, w_k \rangle$, yielding $\lfloor (n-1)/2 \rfloor$ two-dimensional blocks. When n is even, the real vector $v^{(n/2)} = (1, -1, 1, -1, \dots) \in V_{\text{std}}$ is an eigenvector with eigenvalue $\hat{h}(n/2) = n/(\omega^{n/2} - 1) = -n/2 < 0$.

Absolute value. By the evaluation identity $\prod_{k=1}^{n-1} (1 - \omega^k) = n$ (substitute $x = 1$ in $(x^n - 1)/(x - 1)$),

$$|\det(A)| = \prod_{k=1}^{n-1} |\hat{h}(k)| = \frac{n^{n-1}}{\prod_{k=1}^{n-1} |\omega^k - 1|} = \frac{n^{n-1}}{n} = n^{n-2}.$$

Sign. The number of negative eigenvalues is $\lfloor (n-1)/2 \rfloor$ (one per conjugate pair) plus 1 when n is even (from $\hat{h}(n/2) < 0$), totalling $\lfloor n/2 \rfloor$. Hence $\det(A) = (-1)^{\lfloor n/2 \rfloor} n^{n-2}$ and $\det(E_{\text{std}}) = n \det(A) = (-1)^{\lfloor n/2 \rfloor} n^{n-1}$. $\square$

The cyclic determinant formula yields a complete description of the Smith normal form of A (see [12] for general SNF theory) and, consequently, the *cokernel* $K(A) \cong \mathbb{Z}^{n-1}/\text{Im}(A)$ (sometimes called the *critical group* by analogy with graph Laplacians [2]; we avoid the term “sandpile group” since A is not a Laplacian matrix).

Corollary 3.9 (Cokernel of cyclic Latin squares). For the cyclic Latin square of any order $n \geq 2$, the reduced matrix A has Smith normal form $\text{diag}(1, n, n, \dots, n)$ with $n - 2$ copies of n . Equivalently,

$$K(A)_{\text{cyc}} \cong (\mathbb{Z}/n\mathbb{Z})^{n-2}.$$

Proof. Write $A = R + n\Delta$ where $R_{ij} = j + 1$ for all i, j (a rank-one integer matrix) and Δ is an integer matrix; this follows from $A_{ij} = (i+j) \bmod n - (i+n-1) \bmod n \equiv j + 1 \pmod n$.

Step 1 (Divisibility). By multilinearity of the determinant in rows, every $k \times k$ minor of A expands as a sum over subsets S of the k row indices, with each term carrying the factor $n^{|S|}$ from the Δ -rows. Since R has identical rows, any term with $|S| \leq k-2$ (i.e., ≥ 2 rows from R) vanishes. Hence every $k \times k$ minor is divisible by n^{k-1} , so the k -th determinantal divisor $\Delta_k := \gcd$ of all $k \times k$ minors satisfies $n^{k-1} \mid \Delta_k$.

Step 2 (Tightness). Since $\gcd(A_{0,0}, \dots, A_{0,n-2}) = \gcd(1-n, \dots, -1) = 1$, we have $\Delta_1 = 1$. By Proposition 3.8, $|\det A| = n^{n-2}$, so $\Delta_{n-1} = n^{n-2}$.

Step 3 (Pigeonhole). The invariant factors $d_k = \Delta_k/\Delta_{k-1}$ satisfy $d_1 = 1$ and $d_2 \mid d_3 \mid \dots \mid d_{n-1}$ with $\prod_{k=2}^{n-1} d_k = n^{n-2}$. For $k = 2$, Step 1 gives $n \mid \Delta_2 = d_1 d_2 = d_2$, so $d_2 \geq n$. The product bound yields $d_2^{n-2} \leq n^{n-2}$, hence $d_2 = n$. Since $d_3 \mid \dots \mid d_{n-1}$ with product n^{n-3} and each $d_k \geq d_2 = n$, the same squeeze gives $d_k = n$ for all $k \geq 2$ by induction. $\square$

4 The binary criterion: rank and adjugate

In this section we restrict to *even* n and study when the stronger divisibility $n^2 \mid \det(E_{\text{std}})$ holds. The characterization is complete for $n \equiv 2 \pmod{4}$ via the $\mathbb{F}_2$ -rank; for $n \equiv 0 \pmod{4}$, a subtler adjugate criterion resolves the question.

Definition 4.1 (Parity pattern). The *parity pattern* of L is the binary matrix $\mathcal{P} = L \bmod 2 \in \mathbb{F}_2^{n \times n}$. Since each row and column of L is a permutation of $\{1, \dots, n\}$ and n is even, each row and column of $\mathcal{P}$ contains exactly $n/2$ ones and $n/2$ zeros. We call such a matrix *doubly balanced*.

Lemma 4.2. Over $\mathbb{F}_2$:

$$(A \bmod 2)_{ij} = \mathcal{P}_{ij} \oplus \mathcal{P}_{i, n-1}, \quad 0 \leq i, j \leq n-2,$$

where $\oplus$ denotes addition in $\mathbb{F}_2$. In particular, $A \bmod 2$ depends only on $\mathcal{P}$.

Proof. $A_{ij} = L_{ij} - L_{i, n-1}$, and subtraction coincides with addition over $\mathbb{F}_2$. $\square$

Theorem 4.3 ($\mathbb{F}_2$ -rank characterization). Let $n \equiv 2 \pmod{4}$. Then:

$$n^2 \mid \det(E_{\text{std}}) \iff \text{rank}_{\mathbb{F}_2}(A \bmod 2) < n-1.$$

Proof. By Lemma 2.4, $n^2 \mid \det(E_{\text{std}})$ iff $n \mid \det(A)$. Write $n = 2q$ with q odd (since $v_2(n) = 1$).

Odd prime factors. For every odd prime $p \mid n$, $p^{v_p(n)} \mid \det(A)$ holds universally by Remark 3.4. This accounts for the full odd part q of n .

The prime 2. It remains to show $2 \mid \det(A)$. We already know $h \mid \det(A)$ from Theorem 3.2, where $h = n/2 = q$ is odd. Thus $q \mid \det(A)$ and the question reduces to whether $2 \mid \det(A)$:

$$n \mid \det(A) \iff 2 \mid \det(A) \iff \det(A \bmod 2) = 0 \iff \text{rank}_{\mathbb{F}_2}(A \bmod 2) < n-1.$$

The second equivalence uses $\det(A) \bmod 2 = \det(A \bmod 2)$ over $\mathbb{F}_2$. $\square$

Remark 4.4 ($n \equiv 0 \pmod{4}$: the $\mathbb{F}_2$ -rank is necessary but not sufficient). When $n \equiv 0 \pmod{4}$, $h = n/2$ is even. By Lemma 3.1, every row sum of A equals $h \cdot (\text{odd})$, which is even. Hence $A\mathbf{1} \equiv \mathbf{0} \pmod{2}$, i.e. $\mathbf{1} \in \ker_{\mathbb{F}_2}(A \bmod 2)$, so $\text{rank}_{\mathbb{F}_2}(A \bmod 2) \leq n-2$ and $2 \mid \det(A)$ unconditionally. However, for $n^2 \mid \det(E_{\text{std}})$ one needs $2^a \mid \det(A)$ where $a = v_2(n) \geq 2$, and the rank condition provides only one factor of 2 beyond the 2^{a-1} already guaranteed by $h \mid \det(A)$. Indeed, for $n = 8$, approximately 11% of Jacobson–Matthews samples achieve $v_2(\det E_{\text{std}}) = 5 = 2v_2(n) - 1$, so that $\frac{n^2}{2} \mid \det(E_{\text{std}})$ but $n^2 \nmid \det(E_{\text{std}})$. A sufficient condition for $n^2 \mid \det(E_{\text{std}})$ when $n \equiv 0 \pmod{4}$ is $\dim \ker_{\mathbb{F}_2}(A \bmod 2) \geq 2$ (Theorem 5.7); a complete characterization is given by Theorem 4.7.

Corollary 4.5. If $n \equiv 2 \pmod{4}$, then $\mathbf{1} \notin \ker_{\mathbb{F}_2}(A \bmod 2)$. In particular, the kernel mechanism of Remark 4.4 is absent, and whether $n^2 \mid \det(E_{\text{std}})$ depends on the specific Latin square.

Proof. When $n \equiv 2 \pmod{4}$, $h = n/2$ is odd, so each row sum $h \cdot (\text{odd})$ is odd. Hence $(A \bmod 2)\mathbf{1} = \mathbf{1} \neq \mathbf{0}$ over $\mathbb{F}_2$. $\square$

Remark 4.6. For $n \equiv 2 \pmod{4}$, Theorem 4.3 together with Lemma 4.2 shows that the parity pattern $\mathcal{P} = L \bmod 2$ is a *complete invariant* for the n^2 -divisibility question: two Latin squares sharing the same parity pattern either both satisfy $n^2 \mid \det(E_{\text{std}})$, or neither does.

This was verified exhaustively at $n = 6$: among 1252 distinct parity patterns arising from the 9408 reduced Latin squares, every Latin square with a given pattern yields the same $\mathbb{F}_2$ -rank for $A \bmod 2$, with zero exceptions.

We now resolve the case $n \equiv 0 \pmod{4}$.

Theorem 4.7 (Adjugate criterion for $n \equiv 0 \pmod{4}$). Let $n \equiv 0 \pmod{4}$ and set $B = A \bmod 2 \in \mathbb{F}_2^{(n-1) \times (n-1)}$. Then

$$n^2 \mid \det(E_{\text{std}}) \iff \text{adj}(B) \cdot \mathbf{1} = \mathbf{0} \text{ in } \mathbb{F}_2^{n-1}.$$

Proof. By Lemma 2.4, $n^2 \mid \det(E_{\text{std}})$ iff $n \mid \det(A)$. Set $h = n/2$. By Theorem 3.2, $h \mid \det(A)$; write $\det(A) = hq$ with $q \in \mathbb{Z}$. Then $n \mid \det(A)$ iff $2 \mid q$.

Step 1. From the adjugate identity $\det(A) \cdot \mathbf{1} = \text{adj}(A) \cdot A\mathbf{1}$ and $A\mathbf{1} = h\mathbf{c}$ (Lemma 3.1), where $c_i = n + 1 - 2L_{i,n-1}$ is always odd, we obtain

$$q \cdot \mathbf{1} = \text{adj}(A) \cdot \mathbf{c}.$$

Step 2. Reduce modulo 2. Since every c_i is odd, $\mathbf{c} \equiv \mathbf{1} \pmod{2}$. Since the determinant is polynomial in the matrix entries, $\text{adj}(A) \bmod 2 = \text{adj}(A \bmod 2) = \text{adj}(B)$. Hence

$$q \bmod 2 = [\text{adj}(B) \cdot \mathbf{1}]_i \quad \text{for every } i,$$

and $2 \mid q$ iff $\text{adj}(B) \cdot \mathbf{1} = \mathbf{0}$ in $\mathbb{F}_2^{n-1}$. □

Corollary 4.8 (Left-kernel parity criterion). Let $n \equiv 0 \pmod{4}$ and $B = A \bmod 2$ with $\dim \ker_{\mathbb{F}_2}(B) = 1$. Let $\lambda \in \mathbb{F}_2^{n-1}$ generate $\ker(B^T)$. Then

$$n^2 \mid \det(E_{\text{std}}) \iff \text{wt}(\lambda) \equiv 0 \pmod{2}.$$

Proof. Since $\text{rank}(B) = n-2$ and $B\mathbf{1} = \mathbf{0}$ (so $\ker(B) = \text{span}\{\mathbf{1}\}$), every column of $\text{adj}(B)$ lies in $\ker(B) = \text{span}\{\mathbf{1}\}$ and every row in $\ker(B^T) = \text{span}\{\lambda\}$. Since $\text{adj}(B) \neq 0$ (at least one $(n-2) \times (n-2)$ minor is nonzero), we have $\text{adj}(B) = \mathbf{1} \cdot \lambda^T$. Hence $\text{adj}(B) \cdot \mathbf{1} = (\lambda^T \mathbf{1}) \cdot \mathbf{1}$, which vanishes iff $\text{wt}(\lambda)$ is even. The result follows from Theorem 4.7. □

Remark 4.9 (Special cases of the adjugate criterion). (i) If $\dim \ker_{\mathbb{F}_2}(B) \geq 2$, then $\text{adj}(B) = 0$ and the criterion holds automatically, recovering Theorem 5.7.

(ii) As a consistency check, Corollary 4.8 was tested on 347 Jacobson–Matthews samples with $\dim \ker = 1$ at $n = 8, 12, 16$; no misclassifications were found.

Example 4.10. For a Latin square L of order 8 with rows $(4, 3, 6, 8, 5, 7, 2, 1)$, $(2, 7, 4, 1, 6, 8, 3, 5)$, $(8, 5, 3, 4, 1, 6, 7, 2)$, $(5, 1, 7, 3, 2, 4, 6, 8)$, $(3, 8, 2, 6, 7, 5, 1, 4)$, $(7, 6, 8, 2, 4, 1, 5, 3)$, $(6, 2, 1, 5, 8, 3, 4, 7)$, $(1, 4, 5, 7, 3, 2, 8, 6)$, the difference matrix $B = A \bmod 2 \in \mathbb{F}_2^{7 \times 7}$ has $\text{rank}_{\mathbb{F}_2}(B) = 6$, so $\dim \ker_{\mathbb{F}_2}(B) = 1$ and Theorem 4.7 applies. Computing $\text{adj}(B) \cdot \mathbf{1} = \mathbf{1} \neq \mathbf{0}$, the criterion predicts $64 \nmid \det(E_{\text{std}})$. Direct computation confirms: $\det(A) = -245,244$ with $v_2(\det(A)) = 2$, hence $v_2(\det(E_{\text{std}})) = v_2(8 \cdot \det(A)) = 5 < 6 = v_2(64)$.

Corollary 4.11 (Parity pattern completeness). For every even n , the question $n^2 \mid \det(E_{\text{std}})$ depends only on the parity pattern $\mathcal{P} = L \bmod 2$.

Proof. For $n \equiv 2 \pmod{4}$, this follows from Theorem 4.3 and lemma 4.2. For $n \equiv 0 \pmod{4}$, Theorem 4.7 involves only $B = A \bmod 2$, which depends on $\mathcal{P}$ by Lemma 4.2. □

4.1 Universal counterexamples in the 2 mod 4 case

The $\mathbb{F}_2$ -rank characterization (Theorem 4.3) reduces the question of counterexample existence to finding doubly balanced binary matrices with full $\mathbb{F}_2$ -rank. We now prove that such counterexamples exist for every $n \equiv 2 \pmod{4}$, $n \geq 6$.

Proposition 4.12 (Balanced realizability). Let n be even. Every doubly balanced binary matrix $\mathcal{P} \in \mathbb{F}_2^{n \times n}$ is the parity pattern of some Latin square of order n .

Proof. Since every row and column of $\mathcal{P}$ has exactly $n/2$ ones, the bipartite graph G with biadjacency matrix $\mathcal{P}$ is $(n/2)$ -regular. By König's edge-coloring theorem [9], G decomposes into $n/2$ perfect matchings $M_1, \dots, M_{n/2}$. Assign to the edges of M_r the odd symbol $2r - 1$ (for $r = 1, \dots, n/2$). Similarly, the complement $J - \mathcal{P}$ defines an $(n/2)$ -regular bipartite graph that decomposes into $n/2$ perfect matchings $M'_1, \dots, M'_{n/2}$; assign to the edges of M'_s the even symbol $2s$ (for $s = 1, \dots, n/2$). The resulting $n \times n$ array L is a Latin square (each row and column receives exactly one label from each matching), and $L \bmod 2 = \mathcal{P}$. $\square$

Lemma 4.13 (Rank bridge). Let $n \equiv 2 \pmod{4}$, and let $\mathcal{P} \in \mathbb{F}_2^{n \times n}$ be doubly balanced. Define $B \in \mathbb{F}_2^{(n-1) \times (n-1)}$ by $B_{ij} = \mathcal{P}_{ij} \oplus \mathcal{P}_{i, n-1}$ for $0 \leq i, j \leq n-2$. Then

$$\text{rank}_{\mathbb{F}_2}(B) = \text{rank}_{\mathbb{F}_2}(\mathcal{P}) - 1.$$

Proof. Since $n/2$ is odd, every row and column sum of $\mathcal{P}$ equals $n/2 \equiv 1 \pmod{2}$, so $\mathcal{P}\mathbf{1} = \mathbf{1}$ and $\mathbf{1}^T \mathcal{P} = \mathbf{1}^T$ over $\mathbb{F}_2$.

Let $E = \{y \in \mathbb{F}_2^n : \mathbf{1}^T y = 0\}$; then $\mathcal{P}$ maps E to E (since $\mathbf{1}^T(\mathcal{P}y) = \mathbf{1}^T y = 0$ for $y \in E$). Define $T: \mathbb{F}_2^{n-1} \rightarrow \mathbb{F}_2^n$ by $(T(x))_j = x_j$ for $j \leq n-2$ and $(T(x))_{n-1} = \mathbf{1}^T x$ (i.e. the last coordinate is the parity of the first $n-1$). Then T is an isomorphism $\mathbb{F}_2^{n-1} \xrightarrow{\sim} E$. A direct computation shows that $T^{-1}\mathcal{P}T = B$ (the (i, j) -entry of $T^{-1}\mathcal{P}T$ is $\mathcal{P}_{ij} \oplus \mathcal{P}_{i, n-1} = B_{ij}$). Therefore $\text{rank}(B) = \text{rank}(\mathcal{P}|_E)$.

Since $\mathcal{P}\mathbf{1} = \mathbf{1} \neq \mathbf{0}$, the vector $\mathbf{1}$ is not in $\ker(\mathcal{P})$, so $\ker(\mathcal{P})$ is entirely contained in E . Thus $\dim \ker(\mathcal{P}|_E) = \dim \ker(\mathcal{P})$ and

$$\text{rank}(B) = \dim E - \dim \ker(\mathcal{P}) = (n-1) - (n - \text{rank}(\mathcal{P})) = \text{rank}(\mathcal{P}) - 1. \quad \square$$

Theorem 4.14 (Universal counterexamples for $n \equiv 2 \pmod{4}$). For every $n \equiv 2 \pmod{4}$, $n \geq 6$, there exists a Latin square L of order n such that $n^2 \nmid \det(E_{\text{std}})$.

More precisely, let $n = 2k$ with $k \geq 3$ odd, and define

$$S_n = \{0, 1, \dots, k-2, k\} \subseteq \mathbb{Z}_n.$$

Then the circulant parity pattern $\mathcal{P}_{S_n}$ (with $(\mathcal{P}_{S_n})_{ij} = \mathbf{1}_{(j-i) \bmod n \in S_n}$) is doubly balanced, lifts to a Latin square by Proposition 4.12, and satisfies $\text{rank}_{\mathbb{F}_2}(\mathcal{P}_{S_n}) = n$. By Lemma 4.13, $\text{rank}_{\mathbb{F}_2}(A \bmod 2) = n-1$, and by Theorem 4.3, $n^2 \nmid \det(E_{\text{std}})$.

Proof. Since $|S_n| = k = n/2$, the circulant $\mathcal{P}_{S_n}$ has exactly $n/2$ ones in every row and column, so it is doubly balanced. By the standard circulant criterion over $\mathbb{F}_2$, $\mathcal{P}_{S_n}$ is invertible if and only if

$$\gcd(f_n(x), x^n - 1) = 1,$$

where $f_n(x) = \sum_{s \in S_n} x^s = 1 + x + \dots + x^{k-2} + x^k$ is the indicator polynomial. Since $\text{char } \mathbb{F}_2 = 2$ and $n = 2k$,

$$x^n - 1 = x^n + 1 = (x^k + 1)^2,$$

so it suffices to show $\gcd(f_n, x^k + 1) = 1$.

Suppose α is a root of $x^k + 1$ in the algebraic closure $\overline{\mathbb{F}_2}$, i.e. $\alpha^k = 1$.

Case $\alpha = 1$. $f_n(1) = |S_n| = k \equiv 1 \pmod{2}$ (since k is odd), so $f_n(1) \neq 0$.

Case $\alpha \neq 1$. Observe that

$$(x+1)f_n(x) = 1 + x^{k-1} + x^k + x^{k+1},$$

which is verified by expanding $(x+1)(1+x+\dots+x^{k-2}+x^k)$ over $\mathbb{F}_2$. Evaluating at α with $\alpha^k = 1$:

$$(\alpha+1)f_n(\alpha) = 1 + \alpha^{k-1} + \alpha^k + \alpha^{k+1} = 1 + \alpha^{-1} + 1 + \alpha = \alpha^{-1} + \alpha.$$

If $f_n(\alpha) = 0$ then $\alpha^{-1} + \alpha = 0$, i.e. $\alpha^2 = 1$, whence $\alpha = 1$ (the only solution in characteristic 2), contradicting $\alpha \neq 1$.

Therefore $\gcd(f_n, x^k + 1) = 1$, so $\mathcal{P}_{S_n}$ is invertible over $\mathbb{F}_2$, i.e. $\text{rank}_{\mathbb{F}_2}(\mathcal{P}_{S_n}) = n$. By Lemma 4.13, $\text{rank}_{\mathbb{F}_2}(B) = n - 1$ where $B = A \bmod 2$ (Lemma 4.2). Theorem 4.3 then gives $n^2 \nmid \det(E_{\text{std}})$. $\square$

Remark 4.15. The skip-one family S_n was originally identified by exhaustive search over all cyclic subsets $S \subseteq \mathbb{Z}_n$ with $|S| = n/2$, and verified computationally to give full $\mathbb{F}_2$ -rank for all $n \equiv 2 \pmod{4}$ up to $n = 198$ (49 values). The algebraic proof above shows the family works for all such n .

Witness determinant values from one specific lift (via the perfect matching decomposition of Proposition 4.12) are non-canonical: different matching decompositions may produce different Latin squares with the same parity pattern, and hence different determinant values. However, the $\mathbb{F}_2$ -rank—and therefore the counterexample property—depends only on the parity pattern (Corollary 4.11).

Remark 4.16 (Consecutive interval degeneration). The complementary choice $T_n = \{0, 1, \dots, k-1\}$ (a consecutive interval) always yields $\text{rank}_{\mathbb{F}_2}(\mathcal{P}_{T_n}) = n/2$, the minimum possible for a doubly balanced circulant. This is because the indicator polynomial $g(x) = 1 + x + \dots + x^{k-1} = (x^k + 1)/(x + 1)$ divides $x^k + 1$ over $\mathbb{F}_2$, so $\gcd(g, x^{2k} + 1) = (x^k + 1)/(x + 1)$ has degree $k - 1$, and the rank drops by $k - 1$. The skip-one modification $S_n = T_n \setminus \{k - 1\} \cup \{k\}$ breaks this divisibility and restores full rank.

5 Structural properties of $A \bmod 2$

In this section we derive finer structural results about the binary matrix $A \bmod 2$.

Theorem 5.1 (Constant row weight). Let n be even and let $\mathcal{P} = L \bmod 2$ be doubly balanced (i.e. every row and column of $\mathcal{P}$ has exactly $n/2$ ones). Then every row of $A \bmod 2$ has Hamming weight exactly $n/2$.

Proof. By Lemma 4.2, $(A \bmod 2)_{ij} = \mathcal{P}_{ij} \oplus \mathcal{P}_{i, n-1}$ for $0 \leq j \leq n - 2$. Fix a row i . Since $\mathcal{P}$ is doubly balanced, row i of $\mathcal{P}$ has exactly $n/2$ ones among its n entries.

Case $\mathcal{P}_{i, n-1} = 1$. Then $(A \bmod 2)_{ij} = 1$ iff $\mathcal{P}_{ij} = 0$. Among the first $n - 1$ columns, the number of zeros is $(n - 1) - (n/2 - 1) = n/2$ (subtracting the $n/2 - 1$ ones from columns $0, \dots, n - 2$ given that one of the $n/2$ total ones is in column $n - 1$). Hence $\text{wt}(\text{row } i) = n/2$.

Case $\mathcal{P}_{i, n-1} = 0$. Then $(A \bmod 2)_{ij} = 1$ iff $\mathcal{P}_{ij} = 1$. All $n/2$ ones of row i lie in columns $0, \dots, n - 2$. Hence $\text{wt}(\text{row } i) = n/2$. $\square$

Remark 5.2. The column weights of $A \bmod 2$ are *not* constant in general. Column j has weight $|\{i < n - 1 : \mathcal{P}_{ij} \neq \mathcal{P}_{i, n-1}\}|$, which depends on the correlation between columns j and $n - 1$ of $\mathcal{P}$. This asymmetry arises because the definition of A privileges the last column.

The rows of $\mathcal{P}$ form a constant-weight- $(n/2)$ binary code of length n . The following result shows that a simple combinatorial property of this code—the existence of a pair of rows with disjoint supports—forces a rank deficit.

Proposition 5.3 (Disjoint-support obstruction). Let $n \equiv 2 \pmod{4}$ and let $\mathcal{P} \in \{0, 1\}^{n \times n}$ be doubly balanced (all row and column sums equal to $k = n/2$). If $\mathcal{P}$ has two rows with disjoint supports, then $\text{rank}_{\mathbb{F}_2}(\mathcal{P}) \leq n - 1$.

Proof. Since k is odd, each of the n column sums equals $k \equiv 1 \pmod{2}$, so over $\mathbb{F}_2$ the sum of all rows is $\sum_{\ell=1}^n r_\ell = \mathbf{1}$. If rows r_i and r_j have disjoint supports of size k each with $2k = n$, then $r_i + r_j = \mathbf{1}$ over $\mathbb{F}_2$. Substituting: $\sum_{\ell \neq i, j} r_\ell = \mathbf{0}$, which is a nontrivial linear dependency among $n - 2 \geq 4$ rows. Hence $\text{rank}_{\mathbb{F}_2}(\mathcal{P}) \leq n - 1$. $\square$

Remark 5.4. The converse of Proposition 5.3 holds at $n = 6$ but fails for $n \geq 10$; see Proposition 7.2 and remark 7.8 below.

Theorem 5.5 (Odd-prime kernel membership). Let p be an odd prime dividing n . Then $\mathbf{1} \in \ker(A \bmod p)$, and consequently $p \mid \det(A)$.

Proof. By Lemma 3.1, $(A\mathbf{1})_i = \frac{n}{2}(n+1-2L_{i,n-1})$. Since $p \mid n$ and $\gcd(p, 2) = 1$, we have $p \mid \frac{n}{2} \cdot (\dots)$ (indeed $p \mid n$ implies $p \mid (A\mathbf{1})_i$). Thus $A\mathbf{1} \equiv \mathbf{0} \pmod{p}$. By the adjugate identity $\det(A) \cdot \mathbf{1} = \text{adj}(A) \cdot A\mathbf{1}$, it follows that $p \mid \det(A)$. $\square$

Corollary 5.6 (Exact prime structure of the obstruction). The divisibility $n^2 \mid \det(E_{\text{std}})$ can fail *only* at the prime 2. For all odd prime factors of n , the required divisibility is guaranteed unconditionally. When $n \equiv 2 \pmod{4}$, $n^2 \mid \det(E_{\text{std}})$ holds iff $\text{rank}_{\mathbb{F}_2}(A \bmod 2) < n-1$ (Theorem 4.3); this can fail for any such n . When $n \equiv 0 \pmod{4}$, $\text{rank}_{\mathbb{F}_2}(A \bmod 2) < n-1$ holds universally but does not suffice for n^2 -divisibility; the complete criterion is $\text{adj}(A \bmod 2) \cdot \mathbf{1} = \mathbf{0}$ (Theorem 4.7).

Theorem 5.7 (Sufficient condition for $n \equiv 0 \pmod{4}$). Let n be even. If $\dim \ker_{\mathbb{F}_2}(A \bmod 2) \geq 2$, then $n^2 \mid \det(E_{\text{std}})$.

Proof. By Theorem 3.2, $h = n/2$ divides $\det(A)$. Write $\det(A) = hq$ with $q \in \mathbb{Z}$. It suffices to show $2 \mid q$, since then $\det(A) \equiv 0 \pmod{2h} = 0 \pmod{n}$ and $\det(E_{\text{std}}) = n \det(A) \equiv 0 \pmod{n^2}$.

By the adjugate identity applied to $\mathbf{1}$:

$$\det(A) \cdot \mathbf{1} = \text{adj}(A) \cdot A\mathbf{1} = h \text{adj}(A) \cdot \mathbf{c},$$

where $c_i = n+1-2L_{i,n-1}$ are all odd (Lemma 3.1). Hence $q = [\text{adj}(A) \cdot \mathbf{c}]_i$ for any i .

Set $B = A \bmod 2$. Since $\dim \ker_{\mathbb{F}_2}(B) \geq 2$, $\text{rank}_{\mathbb{F}_2}(B) \leq n-3$. Every $(n-2) \times (n-2)$ submatrix of B has rank at most $n-3 < n-2$, so every $(n-2) \times (n-2)$ minor of B vanishes over $\mathbb{F}_2$. Since the entries of $\text{adj}(A)$ are (up to sign) such minors, and $\det(M) \bmod 2 = \det(M \bmod 2)$, we conclude $\text{adj}(A) \equiv 0 \pmod{2}$. Therefore $q = [\text{adj}(A) \cdot \mathbf{c}]_i \equiv 0 \pmod{2}$. $\square$

Remark 5.8. For $n \equiv 0 \pmod{4}$, the condition $\dim \ker_{\mathbb{F}_2}(B) \geq 2$ reduces to: *does $\ker_{\mathbb{F}_2}(A \bmod 2)$ contain a vector linearly independent from $\mathbf{1}$?* This is sufficient but not necessary: at $n = 8$, approximately 73% of Jacobson–Matthews samples with $\dim \ker = 1$ (the minimum) still satisfy $8 \mid \det(A)$. The complete criterion is given by Theorem 4.7.

6 Smith normal form and p -adic bounds

The adjugate argument in the proof of Theorem 5.7 combines with the Smith normal form to yield a sharper result.

Theorem 6.1 (2-adic lower bound). Let n be even and let $k = \dim \ker_{\mathbb{F}_2}(A \bmod 2)$. Then

$$v_2(\det A) \geq v_2(n/2) + \max(0, k-1),$$

where $v_2(m)$ denotes the 2-adic valuation of m .

Proof. If $\det(A) = 0$, then $v_2(\det A) = +\infty$ by convention, so there is nothing to prove. Hence assume $\det(A) \neq 0$, and set $h = n/2$.

Let $d_1 \mid d_2 \mid \dots \mid d_{n-1}$ be the invariant factors (Smith normal form) of A over $\mathbb{Z}$. By standard SNF theory [12], the number of invariant factors divisible by 2 equals $\dim \ker_{\mathbb{F}_2}(A \bmod 2) = k$; these are the last k factors $d_{n-k}, \dots, d_{n-1}$.

Key claim: $v_2(d_{n-1}) \geq v_2(h)$.

Let $\Delta_{n-2} = \gcd$ of all $(n-2) \times (n-2)$ minors of A (the $(n-2)$ -th determinantal divisor), so that $d_{n-1} = \det(A)/\Delta_{n-2}$. The entries of $\text{adj}(A)$ are, up to sign, these $(n-2) \times (n-2)$ minors of A , so $\Delta_{n-2} \mid \text{adj}(A)_{ij}$ for all i, j . Write $\text{adj}(A) = \Delta_{n-2} Q$ with $\gcd(Q_{ij}) = 1$.

From the adjugate identity $\det(A) \cdot \mathbf{1} = \text{adj}(A) \cdot A\mathbf{1} = h \text{adj}(A) \mathbf{c} = h \Delta_{n-2} Q \mathbf{c}$, we obtain

$$d_{n-1} = \frac{\det(A)}{\Delta_{n-2}} = h (Q \mathbf{c})_i \quad \text{for each } i,$$

so $h \mid d_{n-1}$ and $v_2(d_{n-1}) \geq v_2(h)$.

Conclusion. If $k = 0$ then $v_2(\det A) \geq v_2(h) + 0$ is just Theorem 3.2. For $k \geq 1$, the k even factors satisfy $v_2(d_{n-k}) \leq \dots \leq v_2(d_{n-1})$ by the divisibility chain. The last factor contributes $v_2(d_{n-1}) \geq v_2(h)$ and the remaining $k - 1$ factors each contribute $v_2 \geq 1$, giving

$$v_2(\det A) = \sum_{i: 2 \mid d_i} v_2(d_i) \geq v_2(h) + (k - 1). \quad \square$$

Remark 6.2. The bound of Theorem 6.1 is consistent with every observed kernel dimension at $n = 8, 12, 16$: equality is attained at each value of k realized by Jacobson–Matthews sampling.

The proof above uses nothing specific to $p = 2$: the row-sum identity, the adjugate divisibility, and the SNF pigeonhole all hold at every prime. The only subtlety is that, for n odd, the effective row-sum eigenvalue is $h = n$ rather than $h = n/2$ (Lemma 3.1).

Corollary 6.3 (General p -adic lower bound). Let p be any prime dividing n , and let $k_p = \dim \ker_{\mathbb{F}_p}(A \bmod p)$.

- If n is even, then $v_p(\det A) \geq v_p(n/2) + \max(0, k_p - 1)$.
- If n is odd (so p is necessarily odd), then $v_p(\det A) \geq v_p(n) + \max(0, k_p - 1)$.

For odd p the two cases give the same numerical bound $v_p(n) + \max(0, k_p - 1)$, since $v_p(n/2) = v_p(n)$ when $\gcd(p, 2) = 1$. For every odd prime $p \mid n$, we have $k_p \geq 1$ because $A\mathbf{1} \equiv \mathbf{0} \pmod{p}$ by the row-sum identity, so the odd-prime bound becomes stronger than the base divisibility exactly when $k_p \geq 2$. For $p = 2$, the behavior depends on $n \bmod 4$: if $n \equiv 0 \pmod{4}$, then $\mathbf{1} \in \ker_{\mathbb{F}_2}(A \bmod 2)$ by Remark 4.4; if $n \equiv 2 \pmod{4}$, then $k_2 = 0$ is possible and corresponds to the full-rank obstruction of Theorem 4.3.

Proof. If $\det(A) = 0$, then $v_p(\det A) = +\infty$ by convention, so there is nothing to prove. Hence assume $\det(A) \neq 0$.

The argument follows Theorem 6.1 with 2 replaced by p .

Case n even. By Lemma 3.1, $A\mathbf{1} = h \mathbf{c}$ with $h = n/2$ and each c_i odd. The adjugate factorization $d_{n-1} = h (Q \mathbf{c})_i$ gives $v_p(d_{n-1}) \geq v_p(h) = v_p(n/2)$. The SNF pigeonhole over the k_p invariant factors divisible by p yields the stated bound.

Case n odd. By Lemma 3.1, $A\mathbf{1} = n \mathbf{u}$ with $u_i = (n+1-2L_{i,n-1})/2 \in \mathbb{Z}$. The adjugate factorization gives $d_{n-1} = n (Q \mathbf{u})_i$, so $v_p(d_{n-1}) \geq v_p(n)$. The SNF pigeonhole completes the proof as before. $\square$

Remark 6.4. The bound of Corollary 6.3 is consistent with every observed (n, p, k_p) triple: 2700 Jacobson–Matthews samples across $n \leq 20$ and $p \in \{2, 3, 5\}$ produced zero violations, and equality was observed in every sampled case.

7 Counterexamples

We now exhibit explicit instances of the counterexample mechanisms already identified in Section 4. The aim of this section is illustrative: to show the universal theorem at work in concrete families and to record the finite behavior at the first accessible orders.

In particular, we exhibit Latin squares for which $n^2 \nmid \det(E_{\text{std}})$. For $n \equiv 2 \pmod{4}$, Theorem 4.14 guarantees the existence of counterexamples at every order $n \geq 6$; Theorem 4.3 provides the underlying criterion $\text{rank}_{\mathbb{F}_2}(A \bmod 2) = n - 1$. For $n \equiv 0 \pmod{4}$, although $\mathbf{1} \in \ker_{\mathbb{F}_2}(A \bmod 2)$ always holds, the bound $\frac{n^2}{2} \mid \det(E_{\text{std}})$ is still sharp (see $n = 8$ in Remark 4.4); however, $n^2 \mid \det(E_{\text{std}})$ is guaranteed whenever $\dim \ker_{\mathbb{F}_2}(A \bmod 2) \geq 2$ (Theorem 5.7).

7.1 Exhaustive classification at $n=6$

There are exactly 9408 reduced Latin squares of order 6 (first row = first column = $(1, 2, 3, 4, 5, 6)$).

Every Latin square of order 6 is equivalent to a reduced one via row and column permutations, which preserve $\det(E_{\text{std}})$ up to sign.

Proposition 7.1. Among the 9408 reduced Latin squares of order 6:

- (i) 8832 satisfy $36 \mid \det(E_{\text{std}})$, with $\text{rank}_{\mathbb{F}_2}(A \bmod 2) \leq 4$.
- (ii) 576 satisfy $18 \mid \det(E_{\text{std}})$ but $36 \nmid \det(E_{\text{std}})$, with $\text{rank}_{\mathbb{F}_2}(A \bmod 2) = 5$ (full rank).

The 576 counterexamples correspond to 84 out of 1252 distinct parity patterns. No parity pattern gives a mixed outcome: every Latin square with the same pattern produces the same $\mathbb{F}_2$ -rank.

The set $\mathcal{R}(6, 3)$ of all doubly balanced 6×6 binary matrices has exactly 297 200 elements. We classify these by the action of the group $\mathcal{G} = (S_6 \times S_6) \rtimes \langle \tau, \kappa \rangle$ (row permutations, column permutations, transpose τ , and complement $\kappa : P \mapsto \mathbf{1} - P$), all of which preserve $\text{rank}_{\mathbb{F}_2}$.

Proposition 7.2 (Orbit classification at $n = 6$). $\mathcal{R}(6, 3)$ decomposes into exactly 6 orbits under $\mathcal{G}$, of sizes 200, 16 200, 86 400, 43 200, 129 600, and 21 600, with $\mathbb{F}_2$ -ranks 2, 3, 4, 4, 5, 6 respectively. In particular there is a *unique* full-rank orbit. At $n = 6$, a doubly balanced pattern has full $\mathbb{F}_2$ -rank if and only if every pair of rows has nonempty support intersection, i.e. the converse of Proposition 5.3 holds.

Remark 7.3. The unique full-rank representative admits a block-circulant presentation: indexing rows and columns by $(a, b) \in \mathbb{Z}_3 \times \mathbb{Z}_2$, the matrix is $\text{circ}(J_2, I_2, 0)$ over $\mathbb{Z}_3$. The skip-one circulant $\text{circ}(\{0, 1, 3\})$ from Theorem 4.14 lies in this orbit.

Remark 7.4. The 84 full-rank parity patterns are characterized by their column weight profiles. Among all $\binom{5}{3}^5 = 100\,000$ binary 5×5 matrices with constant row weight 3, only 19 440 (19.4%) have full $\mathbb{F}_2$ -rank, and only 84 of these are realized by Latin squares—a selection rate of 0.43%. The Latin square structure is far more restrictive than the weight constraint alone.

7.2 Constructed counterexamples at $n=10$

For $n = 10$, exhaustive enumeration is infeasible ($\sim 2 \times 10^{17}$ reduced Latin squares), so we adopt a constructive approach.

Method. By Theorem 4.14, the circulant pattern $\mathcal{P}_{S_{10}}$ with $S_{10} = \{0, 1, 2, 3, 5\} \subseteq \mathbb{Z}_{10}$ is doubly balanced and has full $\mathbb{F}_2$ -rank. By Proposition 4.12, it lifts to a Latin square. The examples below were found before this theoretical result was available, using the following constructive approach:

1. Generate random *doubly balanced* 10×10 binary matrices $\mathcal{P}$ (5 ones per row and column).
2. Compute the 9×9 matrix $A \bmod 2$ via Lemma 4.2 and check $\text{rank}_{\mathbb{F}_2} = 9$.
3. If full-rank, complete $\mathcal{P}$ to a Latin square by decomposing the bipartite graph into perfect matchings (Proposition 4.12), or alternatively by backtracking with minimum-remaining-values heuristic.
4. Verify $\det(A)$ using exact integer arithmetic (Bareiss algorithm).

Proposition 7.5. (i) Among 4000 doubly balanced 10×10 binary matrices sampled via switch-chain MCMC (50 000 burn-in steps, thinning every 200 steps), 711 (17.8%) have full $\mathbb{F}_2$ -rank. The rank distribution is:

$\text{rank}_{\mathbb{F}_2}$	6	7	8	9	10
Count	5	107	990	2187	711

- (ii) All 711 full-rank patterns were successfully completed to Latin squares via the perfect-matching decomposition of Proposition 4.12.
- (iii) Every resulting Latin square had odd $\det(A)$, confirming $\text{rank}_{\mathbb{F}_2}(A \bmod 2) = 9$ and $100 \nmid \det(E_{\text{std}})$. Thus the corpus yields 711 counterexample instances at $n = 10$ (structural distinctness among these instances has not been verified).

Example 7.6 (Explicit counterexample at $n = 10$). The following is a Latin square of order 10 with $\det(A) = 15\,427\,045$ (odd):

$$L = \begin{pmatrix} 2 & 5 & 6 & 4 & 3 & 8 & 7 & 9 & 10 & 1 \\ 10 & 6 & 2 & 7 & 5 & 1 & 9 & 8 & 4 & 3 \\ 3 & 4 & 1 & 10 & 8 & 5 & 2 & 7 & 9 & 6 \\ 7 & 1 & 10 & 8 & 6 & 9 & 3 & 4 & 2 & 5 \\ 1 & 7 & 3 & 6 & 4 & 10 & 8 & 2 & 5 & 9 \\ 9 & 2 & 5 & 3 & 10 & 6 & 4 & 1 & 8 & 7 \\ 6 & 9 & 8 & 1 & 2 & 7 & 5 & 10 & 3 & 4 \\ 5 & 8 & 7 & 2 & 9 & 4 & 1 & 3 & 6 & 10 \\ 4 & 3 & 9 & 5 & 1 & 2 & 10 & 6 & 7 & 8 \\ 8 & 10 & 4 & 9 & 7 & 3 & 6 & 5 & 1 & 2 \end{pmatrix}$$

The relevant invariants are:

Property	Value
Valid Latin square	✓ (rows and columns are permutations of $\{1, \dots, 10\}$)
Doubly balanced parity	✓ (5 odd + 5 even per row and column)
$\det(A)$	$15\,427\,045 = 5 \times 3\,085\,409$
$v_2(\det(A))$	0 (odd)
$\det(E_{\text{std}}) = 10 \det(A)$	$154\,270\,450$
$50 \mid \det(E_{\text{std}})$?	Yes (Theorem 3.2 confirmed)
$100 \mid \det(E_{\text{std}})$?	No ($\Leftarrow$ counterexample)
$\text{rank}_{\mathbb{F}_2}(A \bmod 2)$	$9 = n - 1$ (full rank)

Remark 7.7 (Abundance). The 711 counterexamples at $n = 10$ arise from MCMC-sampled balanced patterns. All satisfy $5 \mid \det(A)$ (as guaranteed by Theorem 5.5) and $2 \nmid \det(A)$. The lift from full-rank balanced pattern to counterexample Latin square succeeded in 711/711 cases (100%), suggesting that the counterexample property depends entirely on the parity pattern (as proved in Corollary 4.11) and that the lift step introduces no additional obstruction.

Remark 7.8 (Converse failure at $n = 10$). Switch-chain MCMC sampling of 20 000 matrices from $\mathcal{R}(10, 5)$ yields approximately 17.8% full-rank. Every full-rank sample satisfies the no-disjoint-pair condition of Proposition 5.3 (recall = 1.000), but the converse fails: among 12 549 samples with no disjoint row pair, 8 995 are rank-deficient (precision = 0.283). Thus the disjoint-support obstruction is necessary but not sufficient for $n \geq 10$.

8 Computational methods

All computations were performed with exact integer arithmetic to avoid numerical artifacts.

All claims explicitly labeled *exhaustive* were verified by finite exact computation. Sampling data are used only as consistency checks or as sources of witnesses, never as proof of a theorem unless exhaustive enumeration is explicitly stated. The scripts used for determinant, Smith normal form, finite-field rank, adjugate testing, and Jacobson–Matthews sampling are part of the accompanying computational project infrastructure.

- **Determinant computation.** The Bareiss algorithm was used to compute $\det(A)$ over $\mathbb{Z}$ in $O(n^3)$ operations, with exact integer division at each step to prevent coefficient growth [1].
- **$\mathbb{F}_2$ -rank computation.** Gaussian elimination over $\mathbb{F}_2$ in $O(n^3)$ bit operations.
- **Smith Normal Form.** For selected verification, the Smith Normal Form of A over $\mathbb{Z}$ was computed to extract invariant factors and confirm p -adic valuations independently.
- **Latin square generation.** Reduced Latin squares of order 6 were enumerated exhaustively (9408 squares). For orders 8–16, Latin squares were sampled via the Jacobson–Matthews Markov chain [8] with at least $5n^2$ mixing steps per sample; we refer to these as *Jacobson–Matthews samples* rather than claiming exact uniformity. All 576 Latin squares of order 4 were enumerated for exhaustive verification.
- **Balanced binary matrices.** Random doubly balanced $n \times n$ binary matrices (with row and column sums equal to $n/2$) were generated by rejection sampling for small orders. For $n = 10$, exploratory pattern-space samples were drawn via switch-chain MCMC on $\mathcal{R}(n, n/2)$ (swapping 2×2 checkerboard submatrices to preserve all row and column sums); no rigorous mixing-time bound is claimed, and these samples are treated as approximately uniform.
- **Constrained completion.** Given a doubly balanced parity pattern $\mathcal{P}$, Latin square completion was performed by backtracking with the minimum-remaining-values (MRV) heuristic and forward checking. Each cell’s domain is restricted to odd or even values according to $\mathcal{P}$, and assignments are propagated to prune domains in the same row and column.

Remark 8.1. For $n = 6$, the exhaustive enumeration and verification of all 9408 reduced Latin squares completes in under one second. For $n = 10$, the backtracking-lift procedure (pattern generation, rank check, backtracking completion, Bareiss determinant) typically produces a counterexample within several hundred trials, each trial requiring a few seconds.

Code and data availability. All scripts, seeds, and certified datasets used in this paper are available as an open-source Python package (`latin_det`) released under the MIT licence at <https://github.com/aconsciousfractal/Determinant-Divisibility-of-Centered-Latin-Squares>. The repository includes: (i) exact Bareiss determinant, $\mathbb{F}_2$ rank, adjugate and Smith-normal-form routines; (ii) a reproducible random Latin-square sampler and a switch-chain MCMC on $\mathcal{R}(n, n/2)$; (iii) certified datasets for $n \in \{5, 6\}$ exhaustive enumeration, the 10,000-sample $n = 8$ table (seed 0), the 4,000-sample $n = 10$ switch-chain

table (seed 42), the cyclic Smith-normal-form table for $n \leq 12$, the p -adic bound scan for $n \leq 12$, $p \in \{2, 3, 5\}$, and the Example 7.6 witness JSON certificate; and (iv) a standalone script (`scripts/verify_isotopy_destruction.py`) reproducing the isotopy non-invariance experiment of §9. A SHA-256 manifest (`results/certified/SHA256SUMS`) pins the byte-exact contents of every deterministic artefact. A permanent archive of the submission snapshot (tag `v1.0-submission`) will be deposited on Zenodo; the DOI will be added upon acceptance.

9 Discussion

9.1 Summary of results

Theorem 3.2 provides a universal quadratic divisibility bound for $\det(E_{\text{std}})$ over all Latin squares. Theorem 4.3 reduces the question of improved n^2 -divisibility to a binary linear algebra problem when $n \equiv 2 \pmod{4}$. Theorem 5.1 and Theorem 5.5 clarify the precise mechanism: every odd prime factor of n contributes its share unconditionally, while the prime 2 is governed by the $\mathbb{F}_2$ -rank of a matrix determined solely by the parity pattern of L . For $n \equiv 0 \pmod{4}$, the all-ones vector lies automatically in the $\mathbb{F}_2$ -kernel of $A \bmod 2$, but this does not suffice to force n^2 -divisibility. Theorem 4.7 provides the complete criterion: $n^2 \mid \det(E_{\text{std}})$ if and only if $\text{adj}(B) \cdot \mathbf{1} = \mathbf{0}$ in $\mathbb{F}_2^{n-1}$, where $B = A \bmod 2$; Theorem 5.7 is the special case $\dim \ker_{\mathbb{F}_2} \geq 2$ (which forces $\text{adj}(B) = 0$). Theorem 6.1 refines the 2-adic analysis to a tight lower bound: $v_2(\det A) \geq v_2(n/2) + \max(0, k-1)$ where $k = \dim \ker_{\mathbb{F}_2}(A \bmod 2)$, proved via Smith normal form theory and the divisibility chain of invariant factors. The unified bound $\frac{n^2}{2} \mid \det(E_{\text{std}})$ remains sharp at $n = 8$.

The overall picture is as follows:

n	Divisibility	Status
n odd	$n^2 \mid \det(E_{\text{std}})$	Universal (Theorem 3.2)
n even	$\frac{n^2}{2} \mid \det(E_{\text{std}})$	Universal; globally sharp for even orders, but not at $n = 4$ (Theorem 3.2, proposition 3.5, and remark 3.6)
$n \equiv 2 \pmod{4}$	$\frac{n^2}{2} \mid \det(E_{\text{std}})$	iff $\text{rank}_{\mathbb{F}_2}(A \bmod 2) < n-1$ (Theorem 4.3)
$n \equiv 2 \pmod{4}$	$n^2 \nmid \det(E_{\text{std}})$	$\exists$ for all $n \geq 6$ (Theorem 4.14)
$n \equiv 0 \pmod{4}$	$n^2 \mid \det(E_{\text{std}})$	iff $\text{adj}(A \bmod 2) \cdot \mathbf{1} = \mathbf{0}$ (Theorem 4.7)
n even	$v_2(\det A) \geq v_2(n/2) + \max(0, k-1)$	Lower bound (Theorem 6.1), $k = \dim \ker_{\mathbb{F}_2}$
odd $p \mid n$	$v_p(\det A) \geq v_p(n) + \max(0, k_p-1)$	Lower bound (Corollary 6.3), $k_p = \dim \ker_{\mathbb{F}_p}$

9.2 Consequences and related directions

Isotopy non-invariance. The $\mathbb{F}_2$ -rank of $A \bmod 2$ —and hence the counterexample property—is *not* an isotopy invariant. Row and column permutations of L preserve $|\det(E_{\text{std}})|$, but symbol relabeling $L \mapsto \gamma(L)$ changes the parity pattern and can alter the $\mathbb{F}_2$ -rank; compare the broader isotopy/autotopism context in [13]. In particular, reducing the $n = 10$ counterexample of Example 7.6 to standard form (first row = $(1, \dots, 10)$) produces a Latin square with *even* $\det(A)$, i.e. the counterexample property is destroyed. This shows that the divisibility deficit depends on the *labeled* Latin square, not merely on its isotopy class or main class.

Connection to coding theory. By Theorem 5.1, the rows of $A \bmod 2$ form a constant-weight binary code of weight $n/2$ and length $n-1$ (see [10] for the general theory of constant-weight codes). The code is not in general doubly constant-weight: column weights vary depending on the correlation structure of the parity pattern. It would be interesting to characterize which constant-weight configurations arise from Latin squares and how the dimension of the row space is constrained.

The integer invariant $f(L)$. For Latin squares satisfying $n^2 \mid \det(E_{\text{std}})$, the quotient $f(L) = \det(E_{\text{std}})/n^2$ is a well-defined integer. For the cyclic Latin square of odd order n , $f(L) =$

$(-1)^{(n-1)/2}n^{n-3}$. The invariant $f(L)$ is preserved by transposition ($f(L) = f(L^T)$) but not by isotopy in general.

Generalization to odd primes. Corollary 6.3 extends the 2-adic bound of Theorem 6.1 to every prime $p \mid n$: for odd p ,

$$v_p(\det A) \geq v_p(n) + \max(0, k_p - 1), \quad k_p = \dim \ker_{\mathbb{F}_p}(A \bmod p).$$

This strengthens the base divisibility of Theorem 3.2 whenever $k_p \geq 2$. We note that the earlier conjectured bound $v_p(n/p) + \max(0, k_p - 1)$ is valid but sub-optimal: it is weaker by exactly one p -adic unit.

Cokernel and the Latin square graph. Corollary 3.9 shows that the cokernel of the cyclic Latin square has the rigid form $(\mathbb{Z}/n)^{n-2}$. For a general Latin square of order n , the cokernel $K(A)$ exhibits far richer structure: exhaustive enumeration at $n = 5$ yields 6 distinct isomorphism types among 56 reduced squares, and at $n = 6$ there are 377 types among 9408 reduced squares. Related Smith-normal-form computations for combinatorially defined matrices (rook graphs, graph Laplacians) have been carried out in [6, 11] and provide a template for extending the present cokernel analysis.

A natural question is whether $K(A)$ embeds algebraically into the critical group $K(\text{LSG}(n))$ of the associated Latin square graph, which would manifest as divisibility $|K(A)| \mid |K(\text{LSG}(n))|$.

Remark 9.1 (Divisibility by $K(\text{LSG})$). Let $\text{LSG}(n)$ denote the Latin square graph on n^2 vertices (two vertices adjacent iff they share a row, column, or symbol). All Latin square graphs of order n are strongly regular with parameters $(n^2, 3(n-1), n-2, 6)$, hence co-spectral; in particular, $|K(\text{LSG}(n))|$ is constant across all Latin squares of order n (though the group structure may vary). Exhaustive computation gives:

$$|K(\text{LSG}(5))| = 2^{12} \cdot 3^{12} \cdot 5^{22}, \quad |K(\text{LSG}(6))| = 2^{48} \cdot 3^{53}.$$

The divisibility $|K(A)| \mid |K(\text{LSG}(n))|$ holds for all reduced Latin squares at $n = 3$ and $n = 4$, but *fails* at $n = 5$ for 14 of 56 squares (25%) and at $n = 6$ for 5688 of 7840 non-singular squares (72.6%).

In every observed case, the failure occurs precisely when $|K(A)|$ has a prime factor absent from $|K(\text{LSG}(n))|$. For instance, at $n = 5$, the failing orders $|K(A)| \in \{55, 105, 155\}$ contain the “alien” primes 11, 7, and 31 respectively; at $n = 6$ the prime support of $|K(\text{LSG}(6))|$ is $\{2, 3\}$, and any Latin square with $|K(A)|$ divisible by a prime $p \geq 5$ fails.

9.3 Open questions

- (1) **Density of counterexamples.** Several distinct frequencies have been observed. At $n = 6$, the exact counterexample rate among *reduced Latin squares* is $576/9408 \approx 6.1\%$. At $n = 8$, approximately 11% of *Jacobson–Matthews samples* (Latin squares, not patterns) satisfy $n^2 \nmid \det(E_{\text{std}})$ (Remark 3.7). At $n = 10$, the doubly balanced *parity-pattern* corpus of Proposition 7.5 yields $711/4000 \approx 17.8\%$ full-rank patterns. These three rates are not directly comparable: the first is exact over reduced squares, the second is a Latin-square sampling frequency, and the third is a pattern-space frequency. Clarifying the relationship among these measures, and determining the asymptotic counterexample density among all Latin squares as $n \rightarrow \infty$ along even n , remains open.
- (2) **Structural characterization of full-rank parity patterns.** At $n = 6$, Proposition 7.2 shows that full $\mathbb{F}_2$ -rank occurs precisely for the unique orbit of doubly balanced patterns with no disjoint row pair; the converse of Proposition 5.3 holds exactly. For $n \geq 10$ with

$n \equiv 2 \pmod{4}$, the disjoint-pair obstruction remains necessary but is no longer sufficient (Remark 7.8). Find a structural characterization of full-rank parity patterns—readable directly from the binary matrix—that extends beyond $n = 6$.

- (3) **Group-theoretic structure.** Among the 80 reduced Latin squares of order 6 that are Cayley tables of groups, 36 (45%) are counterexamples—a much higher rate than the 6.1% among all reduced Latin squares. What role does the group structure play in determining the $\mathbb{F}_2$ -rank?
- (4) **Prime support of $K(\text{LSG}(n))$.** By Remark 9.1, the prime support of $|K(\text{LSG}(n))|$ is $\{2, 3, 5\}$ at $n = 5$ and $\{2, 3\}$ at $n = 6$. Does the prime support grow or shrink with n ? A complete description of $|K(\text{LSG}(n))|$ would determine exactly which cokernels $K(A)$ can embed into the graph-level critical group.

A Determinant values at $n=5$

For all 56 reduced Latin squares of order 5, $\det(E_{\text{std}})$ is divisible by $25 = 5^2$. The distribution of values is:

$\det(E_{\text{std}})$	Count
−525	6
−400	12
−25	16
25	5
275	2
400	8
525	4
625	1
775	2

In particular, $\det(E_{\text{std}}) \neq 0$ for all 56 squares, so $\text{rank}(E_{\text{std}}) = 4 = n - 1$ (full rank on V_{std}) is universal at $n = 5$.

B Weight structure of $n=10$ counterexamples

A representative sample of eight counterexamples at $n = 10$, drawn from the corpus of Proposition 7.5, exhibits constant row weight $5 = n/2$ in $A \bmod 2$ (as predicted by Theorem 5.1) and variable column weights:

#	$\det(A)$	Column weights of $A \bmod 2$ (sorted)
1	2 606 485	(1, 4, 4, 5, 5, 6, 6, 7, 7)
2	−8 863 025	(3, 3, 4, 5, 5, 6, 6, 6, 7)
3	−25 279 735	(2, 4, 4, 5, 5, 5, 6, 7, 7)
4	−24 821 965	(2, 2, 4, 5, 5, 6, 7, 7, 7)
5	−50 014 675	(2, 4, 4, 5, 5, 5, 6, 7, 7)
6	42 371 805	(3, 4, 4, 5, 5, 5, 6, 6, 7)
7	−11 745 775	(1, 5, 5, 5, 5, 6, 6, 6, 6)
8	5 123 485	(2, 4, 4, 5, 5, 5, 6, 7, 7)

All 711 corpus determinants are divisible by 5 (consistent with Theorem 5.5) and none is divisible by 2.

C Exhaustive column weight analysis at $n=6$

Among the 576 counterexamples at $n = 6$, the column weight distributions of $A \bmod 2$ are:

Column weights (sorted)	Count
(2, 3, 3, 3, 4)	320
(1, 3, 3, 4, 4)	256

All 576 counterexamples have constant row weight $3 = n/2$.

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